

Peijing Xu

| Phone: (669) 240-8521 | Gmail: <mailto:peij.xu0@gmail.com> | Yahoo Email: peijing.xu@yahoo.com | GitHub: <https://github.com/ramrow> |
| LinkedIn: <https://www.linkedin.com/in/peijing-xu-362598297/> | Website: <https://ramrow.github.io/> |
Google Scholar: <https://scholar.google.com/citations?user=gWJJaYAAAAJ&hl=en>

EDUCATION

Rensselaer Polytechnic Institute May 2026
Bachelor of Science (Dual Major): Computer System Engineering & Computer Science GPA: 3.85/4.00
• Relevant Coursework: Machine Learning from Data, Introduction to Electronics, Intro to Engineering Design, Data Mining, Computational Vision
ECE Master of Science at Carnegie Mellon University Expected May 2028

AI/ML BACKGROUND

Picasso Bot: Motion Plan and Tune Sep 2024 – Dec 2024
Russell Sage Professor, Electrical, Computer, and Systems Engineering John Wen https://github.com/ramrow/robot_cartoon_protrait
• Created a program that takes selfie as input and using various python libraries, to create a drawn version of the input image, then generate different trajectories for different parts of the face, designed to be sent to a robot and drawn on surface.
• Converts the trajectories into World Frame, then calculate joint angles of robot arm with inverse kinematics, the portrait is drawn with force control and trapezoidal velocity profile at 4 milliseconds per point.
Foam-GPT May 2025 – Present
Assistant Professor Shaowu Pan https://github.com/ramrow/Finetune_research_RPI2025
• Optimized LLMs like llama, Qwen3, Mistral, and gpt-oss through SFT so they would generate OpenFOAM files as the input writer AI for an open-sourced version of Foam-Agent, a multi-agent framework designed to automate complex CFD simulations.
• Gained valuable hands-on experience performing inference and deploying LLMs such as Qwen3.5, Qwen3, and gpt-oss. Utilized several finetuning frameworks transformers, Llama-factory, and Unsloth on Linux GPUs. Learned quantization methods to run 100B plus LLMs on H100s.
• Second author on Paper: Foam-GPT: Fine-Tuning Large Language Model for Agentic Automation of CFD Simulations with Open-FOAM published to Machine Learning and the Physical Sciences Workshop, NeurIPS 2025.
• Paper Link: https://ml4physicsciences.github.io/2025/files/NeurIPS_ML4PS_2025_271.pdf
Computational Vision Spring 2026
• Developed a Region-based Convolutional Neural Network, R-CNN, using PyTorch with a ResNet-18 backbone, implementing custom classification and bounding box regression layers, alongside Intersection over Union and non-maximum suppression algorithms for robust object detection.
• Engineered a feature-matching and image alignment pipeline utilizing OpenCV to extract SIFT key points and descriptors, enabling the computation of fundamental matrices and the visualization of epipolar lines across pairwise image sets.
• Built optimized PyTorch data loaders and evaluation pipelines for multi-class image classification tasks assessing model performance through Mean Average Precision, per-class accuracy, and confusion matrices.

EXPERIENCES

Research and Development Intern Remote
Harp Research Inc. Jan 2025 – May 2025
• Created a portfolio optimization model based on a pre-built quantum FPGA that creates two Qubos from historical stock returns, the current return of stock if sold, a targeted sum, and granularity k and solve it through simulated bifurcation to calculate the most optimized allocations of stock that would maximize profit for a user.
• Conducted extensive research and testing on products from quantum involved companies such as Fixstars with its Amplify library, and Pasqal's Qadence and Pulsar libraries to see if their technology is worth building from.
Undergraduate Student Mentor In-person
Rensselaer Polytechnic Institute Aug 2025 - Present
• Student assistant for Principles of Software, assisted in mentoring around 30 students during the recitation time block. Graded students work from recitation quizzes to homework assignments.

PROJECTS

Embedded Control Fall 2023
• Utilize both a compass and a range sensor on a TI-RSLK Robotic Car for it to automatically rotate to a desired direction and check if the car should move forward or backward based on a predetermined distance to prevent any collisions that the car may experience.
• Implemented pulse width modulation on a TI-RSLK Robotic Car for it to follow track field pattern of any size, from tabletop to room sized.
Electric Circuits: Solar-Powered AM Radio Receiver Spring 2024
• Designed a solar-powered AM radio receiver utilizing LTspice for circuit simulation, incorporating a diode-based power management system that autonomously switches between wall and solar power by selecting the highest available voltage.
• Engineered the RF filtering and demodulation pipeline using a second-order LC bandpass filter tuned to a local 810 kHz AM station, followed by a first-order RC low-pass envelope detector to isolate the audio signal.
• Developed an audio output stage combining a non-inverting operational amplifier for voltage gain and a Class B push-pull transistor amplifier for current gain to drive an 8-ohm speaker.
Principles of Software Summer 2024
• Created an Abstract Data type of graph with edges and nodes using generics in Java and used path finding algorithm (BFS, Dijkstra's) for the significant buildings at Rensselaer Polytechnic Institute to find the shortest distance path from one building to another.
• Created a Graphical User Interface using JavaFX that allows the user to input two building names and then display the path physically on an image of the map of the Rensselaer Polytechnic Institute campus in the user interface.
Multi-Disciplinary Capstone: Aircraft Robotic Assembly Sponsored by Boeing Fall 2025
• Engineered the autonomous navigation and pathfinding subsystem using an A* search algorithm to process real-time LiDAR data, dynamically avoid obstacles, and calculate optimal routes within a simulated manufacturing environment.
• Designed a reactive collision-prevention protocol and a 'return-to-home' feature that utilizes stored wheel RPM, travel duration, and compass headings to accurately reverse-navigate the robotic base to its initial starting point.

TECHNOLOGIES/TOOLS

Programming Languages: C/C++, Python, Java, JavaScript, MATLAB, markdown, Prolog, LaTeX, Haskell, SQL

Skills: Circuit design and analysis, Data structures, Finetune, Data analysis and visualization, Team collaboration, Operating systems (Linux, Windows), Agile Framework: Scrum, understanding of neural networks, Lambda AI, csv, Microsoft Notes, Excel, Word, openclaw, Copilot, Huggingface, Slurm script

Tools:

- CS: Llama-Factory, LangChain, LangGraph, transformers, unsloth, git, torch, NumPy, Pandas, matplotlib, vllm, ollama, GPT, claude, openCV
- Engineering: LTspice, CAD, ALICE, Scopy, Arduino IDE, RobotStudio, Raspberry Pi 4/5